# Zaawansowane techniki programowania

# Laboratorium 8

# ZMIENDAK WIKTOR

**ZADANIA OBOWIĄZKOWE:**

1. Uzupełniamy ShopServiceImpl.createProduct tak aby tworzyć nową encję z danych requestu.

```java
@Override
public ProductEntity createProduct(ProductCreateRequest req) {
  ProductEntity p = new ProductEntity(id: null, req.getName(), req.getPrice(), req.getStock());
  return productRepository.save(p);
}
```

2. Uzupełniamy ShopServiceImpl.getProduct tak aby zwracało encję

```java
@Override
public ProductEntity getProduct(long id) {
  return productRepository.findById(id).orElseThrow(() -> notFound(msg: "Product not found"));
}
```

3. Uzupełniamy ShopServiceImpl.createOrder tak aby walidowało ilość produktów, zmniejszało ich ilość oraz zapisuwało encje produktu.

```java
@Override
@Transactional
public OrderEntity createOrder(OrderCreateRequest req) {
  ProductEntity product = productRepository.findById(req.getProductId())
      .orElseThrow(() -> notFound(msg: "Product not found"));

  int quantity = req.getQuantity();
  if (quantity > product.getStock()) {
    throw badRequest(msg: "Insufficient stock");
  }

  product.setStock(product.getStock() - quantity);
  productRepository.save(product);

  BigDecimal total = product.getPrice().multiply(BigDecimal.valueOf(quantity));
  OrderEntity order = new OrderEntity(id: null, product.getId(), quantity, total, Instant.now());
  return orderRepository.save(order);
}
```

4. Uzupełniamy ShopServiceImpl.getOrder tak aby zwracało zamówienie

```java
@Override
public OrderEntity getOrder(long id) {
  return orderRepository.findById(id).orElseThrow(() -> notFound(msg: "Order not found"));
}
```

## ZADANIA DODATKOWE:

1. Uzupełniamy ShopServiceImpl.searchProducts.

```java
@Override
public Page<ProductEntity> searchProducts(String q, BigDecimal minPrice, BigDecimal maxPrice, Pageable pageable) {
  String namePart = q == null ? "" : q;
  BigDecimal min = minPrice == null ? BigDecimal.valueOf(val: 0) : minPrice;
  BigDecimal max = maxPrice == null ? new BigDecimal(val: "9999999999.99") : maxPrice;
  return productRepository.findByNameContainingIgnoreCaseAndPriceBetween(namePart, min, max, pageable);
}
```

2. Uzupełniamy ShopServiceImpl.listOrders.

```java
@Override
public Page<OrderEntity> listOrders(Pageable pageable) {
  return orderRepository.findAll(pageable);
}
```

## WYNIKI TESTÓW

```
2026-05-04T20:46:11.477+02:00  INFO 10912 --- [          main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-05-04T20:46:11.532+02:00  WARN 10912 --- [          main] JpaBaseConfiguration$JpaWebConfiguration : spring.jpa.open-in-view is enabled by default. Therefore, database queries may
ng. Explicitly configure spring.jpa.open-in-view to disable this warning
2026-05-04T20:46:11.629+02:00  INFO 10912 --- [          main] o.s.b.t.m.w.SpringBootMockServletContext : Initializing Spring TestDispatcherServlet ''
2026-05-04T20:46:11.629+02:00  INFO 10912 --- [          main] o.s.t.web.servlet.TestDispatcherServlet  : Initializing Servlet ''
2026-05-04T20:46:11.630+02:00  INFO 10912 --- [          main] o.s.t.web.servlet.TestDispatcherServlet  : Completed initialization in 1 ms
2026-05-04T20:46:11.635+02:00  INFO 10912 --- [          main] pl.example.lab8.Grade3JpaBasicsTest      : Started Grade3JpaBasicsTest in 0.657 seconds (process running for 16.464)
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 15.90 s -- in pl.example.lab8.Grade3JpaBasicsTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  21.629 s
[INFO] Finished at: 2026-05-04T20:46:11+02:00
[INFO] ------------------------------------------------------------------------
PS C:\Users\ziggs\Desktop\Studies\Cybersecurity\Semestr_I\ZTP\Zmiendak_Wiktor_LAB8\Zmiendak_Wiktor_LAB8>
```

```
2026-05-04T20:46:46.508+02:00  INFO 20104 --- [          main] o.s.b.t.m.w.SpringBootMockServletContext : Initializing Spring TestDispatcherServlet ''
2026-05-04T20:46:46.508+02:00  INFO 20104 --- [          main] o.s.t.web.servlet.TestDispatcherServlet  : Initializing Servlet ''
2026-05-04T20:46:46.509+02:00  INFO 20104 --- [          main] o.s.t.web.servlet.TestDispatcherServlet  : Completed initialization in 0 ms
2026-05-04T20:46:46.514+02:00  INFO 20104 --- [          main] p.example.lab8.Grade4QueryAndPagingTest  : Started Grade4QueryAndPagingTest in 0.491 seconds (process running for 12.
2026-05-04T20:46:46.600+02:00  WARN 20104 --- [          main] PageModule$PlainPageSerializationWarning : Serializing PageImpl instances as-is is not supported, meaning that there
f the resulting JSON structure!
        For a stable JSON structure, please use Spring Data's PagedModel (globally via @EnableSpringDataWebSupport(pageSerializationMode = VIA_DTO))
        or Spring HATEOAS and Spring Data's PagedResourcesAssembler as documented in https://docs.spring.io/spring-data/commons/reference/repositories/core-extensions.html#core.web.

[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 12.09 s -- in pl.example.lab8.Grade4QueryAndPagingTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  15.540 s
[INFO] Finished at: 2026-05-04T20:46:46+02:00
[INFO] ------------------------------------------------------------------------
PS C:\Users\ziggs\Desktop\Studies\Cybersecurity\Semestr_I\ZTP\Zmiendak_Wiktor_LAB8\Zmiendak_Wiktor_LAB8>
```